

LIRUI WANG

5000 25th Ave NE, Seattle, WA 98105
(206) 483-8515 ◊ liruiw@uw.edu ◊ liruiw.github.io

EDUCATION

University of Washington, Seattle, WA

Sep 2016 - Present

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

GPA: 3.97/4.00, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection *March 2019 - Present*

- Implement a network that combines data-driven and traditional method to synthesize grasp pose.
- Prototype RL, IL, and MPC with PyBullet environment to map observed points directly to 6D pose increments in task space.
- Introduce an integrate planner that models joint motion and grasp planning as an optimization problem.
- Propose an online learning framework and a grasp synthesis approach to generate grasp and select grasp online.

6D Object Pose Tracking with the Deep Iterative Matching Network *June 2018 - March 2019*

- Tested DeepIM on YCB datasets and achieved robust results trained on synthetic data only.
- Improved network training stability, pose distribution and occlusion generation.
- Extended DeepIM in real-time tracking for occluded scene in YCB and built AR demos.
- Built renderer and kinematics tree for vision system application on robotics.

Unsupervised Flow and Depth Methods Study

March 2018 - June 2018

- Investigated a few deep learning methods for 3D geometry tasks such as depths, flow, and egomotion.
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a robotic setting.

Baxter Hand-Eye Coordination

Aug 2017 - March 2018

- Worked on a manipulation task with PoseCNN, which enables Baxter to move daily objects.
- Studied on camera models, hand-eye calibration with depth camera, robot kinematics, 3D geometry and ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA

Sep 2019 - Present

Computer Vision Teaching Assistant, UW CSE 455

Shenzhen, China

Jun 2017 - Aug 2017

Robotics Engineer Intern, DJI Technology

- Worked on autonomous robot battle which uses vehicles with TX1 onboard and DJI Manifold drones with Guidance

- Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate
- Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot

LEADERSHIP AND INVOLVEMENT

Seattle, WA

Sep 2016 - Feb 2018

Computer Vision Lead

Advanced Robotics at the University of Washington

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI.
- Performed system integration of controls, navigation and serial communication

PERSONAL PROJECTS

- **Sound Scape:** Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle
- **Music Editor:** Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded 1st place out of 35 teams in Seattle

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

COURSES

- **Computer Science:** Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robustness in Machine Learning (Grad)
- **Electrical Engineering:** Digital Signal Processing, Control Theory, Control Capstones, Embedded System
- **Math:** Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis (Grad), High Dimensional Statistics and Statistical Learning (Grad), Statistical Learning: Modeling, Prediction, And Computing (Grad)
- **English:** Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Systems:** ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32
- **Languages:** Python, C++/C, Matlab, Java
- **Tools:** Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt, Bullet, VTK,
- **Hardware:** 3D Printing, Laser Cutting, Machining, Soldering

EXTRACURRICULARS

- **Mens Pickup Soccer Team:** Played regularly at pickup soccer and joined the intramural soccer competition.
- **Math Club:** Joined the meetings to solve problems with other students in the group.